

MINI PROJECT – I
SYNOPSIS REPORT

Face Authentication based Attendance System

by

Name- Piyush Kumar

Entry No -23bcs061

Under the Guidance

Of

Dr. Rohit Tanwar

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SHRI MATA VAISHNO DEVI UNIVERSITY, KATRA

(School of Computer Science & Engineering)

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Content of Synopsis Report

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1.Title of the Project

Face Authentication based Attendance System

2. Introduction and Background of the Study

A face Authentication based attendance system is an advance method for recording attendance using Machine Learning. It automatically identify and verify attendance through their facial recognition features, eliminating the need for manual attendance marking methods like roll number, ID card etc.

Face Authentication based Attendance Systems use computer vision, machine learning algorithms such as KNN and OpenCV Library for real time face detection and recognition.

3. Objectives of the Project

The main objectives of the project is generally focus on taking attendance by using face authentication. ensuring accuracy, security, time efficiency, and prevention of proxy attendance.

4. Problem Statement / Research Question

We will mark attendance by using face detection and store it in a CSV file for our record during implementation of the model.

5. Methodology / Approach

Step	Methodology / Approach	Description
1	Image Acquisition	Capture student images through a webcam in real-time.
2	Grayscale Conversion	Converts RGB image to grayscale to reduce computational complexity while retaining important features.
3	Face Detection	Use algorithms like Haar Cascade detectors to locate faces in the frame.
5	Face Recognition / Authentication	Compare extracted features with trained data that is stored in pickle file to identify and authenticate the individual.

Step	Methodology / Approach	Description
6	Attendance Marking	If authentication succeeds, mark the attendance record in the system as CSV file.

6. Tools, or Technologies to be Used

A. Programming Language:

Python: Most commonly used language for computer vision and machine learning tasks.

B. Libraries and Modules:

1.OpenCV: An open-source computer vision library often used for image processing and face detection tasks.

2.pickle: pickle is a standard Python module for serializing and deserializing Python objects.

3. OS: The os module provides a way to interact with the operating system.

4.numpy: numpy is a fundamental package for numerical computing in Python. Etc.

7. Expected Outcomes / Deliverables

Outcome: A fully functional Face authentication based Attendance System that can mark attendance in a separate file by verifying individual faces from testing data.

8. Timeline of Work (in phases along with completion dates of each task)

Phase-1

SR No	Task	Description	Completion Date
1	Technology Selection	Import necessary libraries and Modules like OpenCV, Haar Cascade (for detection), OS, numpy etc.	10 Sept 2025
2	Data Collection & Preprocessing	Capture images of users, preprocess (grayscale, resize).	15 Sept 2025
3	Face Detection Module	Implement Haar Cascade face detection with detectMultiScale().	18 Sept 2025
4	Feature Extraction	Extract features and save it into .pkl file.	22 Sept 2025

Phase-2

5	Training The Model	Apply ML Algorithm KNN for training the model.	10 Oct 2025
6	Face Recognition	Compare extracted features with trained data that is stored in pickle file to identify and authenticate the individual.	30 Oct 2025
7	Attendance Marking	Create database(CSV) to store attendance records automatically.	20 Nov 2025

9. References (if any)

1. Blogs related to this project (computer vision)
- 2.Documentation regarding different libraries and modules.
3. YouTube videos